

Vishnu Pratap Singh Parihar

Research Scholar | Mehta Family School of Data Science and Artificial Intelligence

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SUMMARY

I am a researcher in **applied mathematics and machine learning**, focused on building **efficient, reliable, and trustworthy AI systems** under **constraints, uncertainty, and real-world variability**. I am currently a PhD student in the **Mehta Family School of Data Science and Artificial Intelligence** at the **Indian Institute of Technology Roorkee**, where I study **risk-aware and distribution-sensitive learning frameworks** for neural networks. My research addresses a fundamental limitation of modern AI: models optimized for average-case performance often fail unpredictably across domains and contexts. To address this, I develop principled methods grounded in **statistics, mathematical analysis, and optimization** to quantify, control, and mitigate **worst-case risks** in model compression and adaptation. My recent work on **Risk-Aware Vetting (RAV)** (*Manuscript under review at a top-tier computer vision venue, 2026*) formulates structural model reduction as a **risk-sensitive decision problem**, using **tail-risk criteria** to prevent hidden failures under domain heterogeneity. In parallel, I investigate **dynamic token adaptation in large multimodal models (LLaVA-7B, LLaVA-13B, and LLaVA-NeXT)**, proposing **decode-aware** mechanisms for efficient and context-sensitive token pruning. Previously, I completed my **M.Sc. in Applied Mathematics from IIT Mandi**, after qualifying **IIT JAM in Mathematics** for postgraduate study. Following my master's, I qualified **GATE Mathematics**, which further strengthened my transition toward advanced research. I also worked as a **Junior Research Fellow at Naval Science and Technological Laboratory (NSTL) DRDO**, where I developed a strong foundation in applied research and scientific computing. Alongside my academic journey, I have taught and mentored 10+2 students in my Graduation and, after my master's, coached students preparing for **NDA** and **IIT-JEE Main** examinations. This sustained engagement with teaching has continually refined my ability to translate deep mathematical ideas into accessible intuition, structured reasoning, and effective problem-solving strategies. Broadly, my research vision is to advance **efficient, reliable, and green AI**, where performance is measured not only by accuracy, but also by **stability, fairness, and trustworthiness** across diverse real-world environments.

EDUCATION

PhD in Data Science & Artificial Intelligence, IIT Roorkee

- ✦ **Thesis:** Risk-Aware Efficient Deep Learning for Reliable Visual Systems (*Ongoing*)
- ✦ **Advisors:** Prof. Sanjeev Kumar (HOD, DS&AI); Prof. Balasubramanian Raman (HOD, CSE)
- ✦ **Research Area:** Efficient and Reliable Deep Learning for Computer Vision
- ✦ **Duration:** July 2023 – Present

MSc in Applied Mathematics, IIT Mandi

- ✦ **Thesis:** Nonlinear dynamics in the study of birdsong
- ✦ **Advisors:** Prof. Padmanabhan Rajan (EE); Prof. Nitu Kumari (Mathematics)
- ✦ **Focus:** Applied Mathematics
- ✦ **Duration:** 2017 – 2019

BSc in Mathematics & Statistics, CSJMU Kanpur

- ✦ **Merit:** Qualified IIT-JAM Mathematics

HONORS & AWARDS

National Fellowships

- ✦ **Junior Research Fellowship (JRF)** — DRDO, Ministry of Defence, India (2022)
- ✦ **Institute Assistantship** — IIT Roorkee, Ministry of Education, India (2023)

National Exams (Mathematics)

- ♦ All India Rank 576, 808 — GATE-22, 23^[1]
- ♦ All India Rank 313 — JAM-17^[2]
- ♦ Post-Graduate Merit — Successfully cleared BHU and DU M.Sc. Entrance Exams.

Other nation-wide exams for which I cleared the written exam but failed to make the final merit list: UPSC National Defence Academy (NDA) 2013 written exams (II), UPSC National Defence Academy (NDA) 2014 written exams (I), TES-2014 to join Indian Army as an Officer and CES-2014 to join Indian Navy as an Officer. I qualified to appear in the Service Selection Board interview 5 times and was conferenced out each time^[3]. **PhD Selection Merit** — Shortlisted for interviews at IIT Bombay (C-MINDS), IIT Delhi (Maths), and IIT Guwahati (Maths) after qualifying highly competitive national-level written entrance examinations.

[1] GATE is a national examination taken annually by 800,000+ candidates, a gateway to postgraduate/doctoral programs in India’s premier institutions.
[2] JAM is a selective national entrance exam (~70,000 candidates/year) for advanced studies in the pure sciences at India’s premier institutions.
[3] SSB is a five-day Interview Evaluation Process to select Defence Force Officers; approximately 90% of called candidates are filtered on the first day itself, while the remaining candidates are evaluated over the full five days.

RESEARCH EXPERIENCE

Doctoral Researcher July 2023 – Present
IIT Roorkee, Mehta Family School of DS & AI Roorkee, India

- **Risk-Aware Model Compression:** Developed the *RAV (Risk-Aware Vetting)* framework to optimize the efficiency-reliability trade-off. Formulated structural pruning as a risk-sensitive decision problem using *Conditional Value-at-Risk (CVaR)* to mitigate “tail-risk” failures.
- **Multimodal Efficiency (VLM):** Investigating *DynaPruMerge* for dynamic visual token reduction in *LLaVA* architectures; introducing decode-aware mechanisms to preserve critical visual evidence in generative vision-language tasks.
- **High-Performance Computing:** Leveraging large-scale GPU clusters *ParamGanga* and *departmental cluster nodes* for training and vetting Vision-Language Models (VLMs) with multi-billion parameter scales.

Junior Research Fellow (JRF) Nov 2022 – June 2023
DRDO, Naval Science & Technological Laboratory (NSTL) Visakhapatnam, India

- **Maritime Field Operations:** Selected as a core team member for the on-board maritime trials of the indigenous heavyweight torpedo *Varunastra*. Collaborated with a multidisciplinary team of *DRDO Scientists, BDL Engineers, and Indian Navy officials* to conduct real-time system monitoring and data acquisition during the *June 2023 live-warhead testing*; a milestone mission featured in national news.
- **Systems Engineering Collaboration:** Investigated the *Weapon Target Allocation (WTA)* problem in coordination with the *System Engineering Division*; contributed to mathematical modeling and statistical trajectory estimation for underwater tactical scenarios.
- **Field Data Analytics:** Executed rigorous data validation and performance analysis during high-stakes sea trials, ensuring autonomous system reliability under extreme underwater operational constraints.

M.Sc. Research July 2018 – May 2019
Indian Institute of Technology (IIT) Mandi Mandi, India

- **Nonlinear Dynamics:** Investigated chaotic phenomena in *biological vocalizations* through mathematical modeling and time-series analysis of avian acoustic signals.
- **Computational Modeling:** Utilized differential equations and bifurcation analysis to study the stability and complex dynamics of biological systems.

PUBLICATIONS & MANUSCRIPTS

Conference Manuscripts (AI / Computer Vision)

1. **V. P. S. Parihar**, Sanjeev Kumar, and Balasubramanian Raman, “RAV: Risk-Aware Vetting for Layer Collapse under Heterogeneous Domain Shifts”, *Manuscript under review at a top-tier computer vision venue*, 2026.
Key Contribution: Introduced Selective Domain Collapse (SDC) and a tail-risk-aware framework (RAV) for reliable structural pruning. Validated across CNNs and Vision Transformers (ViT) on multiple domain-shift benchmarks (PACS, OfficeHome).
2. **V. P. S. Parihar**, Sanjeev Kumar, and Balasubramanian Raman, “DynaPruMerge: Decode-Aware Dynamic Visual Token Reduction in Large Multimodal Models”, *Manuscript in Preparation*, 2026.

Technical Reports & Thesis

1. **V. P. S. Parihar** and Systems Engineering Group (NSTL), “Advanced Statistical Trajectory Estimation and Multi-target Allocation Frameworks”, *Internal Technical Report, DRDO (Classified/Non-public)*, 2023.
2. **V. P. S. Parihar**, “Nonlinear dynamics in the study of birdsong”, *M.Sc. Thesis, IIT Mandi*, 2019.

RELEVANT COURSEWORK

- ◇ **Ph.D. Level (IIT Roorkee)**: Computer Vision, Machine Learning, Reinforcement Learning, Statistical Inference, Evolutionary Algorithms, Time Series Analysis, Mathematical Foundations of Data Science.
- ◇ **Open Courseware Training**: *TinyML & Efficient Deep Learning Computing* (MIT 6.5940); *Deep Learning & Deep Learning for Computer Vision* (Stanford CS230/231n); *Advanced Computer Vision* (UCF CAP6412); *Mathematical Foundations of Generative Models* (IISc, Prof. Paritosh Ap).
- ◇ **Graduate Level (M.Sc. IIT Mandi)**: Linear Algebra, Probability & Statistics, Linear Programming, Financial Mathematical Modelling, Modelling and Simulation, Real & Complex Analysis, Dynamical Systems, Ordinary and Partial Differential Equations, Computational Fluid Dynamics, Numerical Analysis, Topology & Functional Analysis.

TEACHING & MENTORSHIP

- ◇ **Graduate Research Mentorship**, *MFSDSAI, IIT Roorkee* (Feb 2026–Present): Assisting my supervisor in supervising M.Tech and B.Tech cohorts by building foundational rigor in **Deep Learning, Transformers, and ViTs**; mentoring M.Tech thesis (MTP) towards **Efficient Vision-Language Models (VLMs)** and guiding B.Tech students through **Mathematics for Machine Learning** (NPTEL, Prof. Pratos AP, IISc) via structured weekly reviews and technical presentations.
- ◇ **Graduate Teaching Assistant**, *MFSDSAI, IIT Roorkee* (July 2024–Present): Taught *Programming in Python (DAC104)*, *Discrete Structures (DAC201)*, *Data Structures & Algorithms (DAC513)*, and *Data Science (DAI101)*; conducting weekly tutorial sessions and technical evaluations.
- ◇ **Tutoring & Competitive Coaching**, *Kanpur & Remote* (2014–2022): Mentored 10+2 students in **Mathematics and Physics** during graduation; provided remote coaching for **NDA (Mathematics)** and **IIT-JEE Main** (2020–2022), translating complex calculus into intuitive problem-solving strategies.

WORKSHOPS & CONFERENCES

- ◇ *Human Perception and Image Analysis in the Era of Machine Learning*: SPARC Course, **IISc Bangalore** (Feb 2025).
- ◇ *International Conference on Computations and Data Science (CoDS-2024)*: **IIT Roorkee** (March 2024).
- ◇ *Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*: **IIT Ropar** (Dec 2023).
- ◇ *International Conference on Soft Computing for Problem Solving (SocProS 2023)*: **IIT Roorkee** (Aug 2023).
- ◇ *Online International Workshop on Machine Learning Applications to Images, IoT and WSN*: **University of Essex & IIT Lucknow** (July 2020).
- ◇ *Finite Element Method for Engineers and Researchers (FEMER-2018)*: Short Term Course, **IIT Mandi** (June 18–22, 2018).
- ◇ *Science Academies Lecture Workshop on Essence of Partial Differential Equations*: **IIT Mandi** (April 2–4, 2018).

TRAVEL GRANTS

- ◇ *IISc Bangalore*: Full travel and subsistence support for the SPARC course on Human Perception and Image Analysis in the Era of Machine Learning (2025).
- ◇ *IIT Guwahati*: Travel grant awarded for appearing in the Ph.D. interview in Mathematics following national-level shortlisting (2022).
- ◇ *SSB*: Travel allowance (AC 3-tier) and full boarding/lodging provided for the Officer-cadre selection process (2013).

TECHNICAL SKILLS

- ◇ *Programming Languages*: Python, C, Matlab
- ◇ *Frameworks*: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers
- ◇ *Tools*: Colab, VS Code, Jupyter, Docker, Slurm, Overleaf
- ◇ *Cloud / Compute*: GCP, HPC clusters
- ◇ *Document Preparation*: $\text{\LaTeX}$, PowerPoint, Markdown
- ◇ *Platforms*: GitHub, Kaggle
- ◇ *Operating Systems*: Linux, Windows

LEADERSHIP & EXTRACURRICULAR

- ◇ *Institutional Responsibility*: Career and Placement Cell Representative at **IIT Mandi** (2018–19); served as **Class Treasurer** (2011), a student-body liaison for discipline and event execution during senior secondary schooling.
- ◇ *Social & Community Leadership*: Active member of NSS at both **IIT Mandi** and **IIT Roorkee**; part of *Student Mess Committee* (2025–Present) at Cautley Bhawan, IIT Roorkee, overseeing logistics and quality management.
- ◇ *Athletics & Sports*: **Bronze Medalist** in Javelin Throw (Ranneeti, IIT Mandi, 2017); represented school in Javelin (Inter-School Athletic Meet, 2009) and served as wicketkeeper-batsman for the school cricket team.
- ◇ *Travel & Cultural Engagement*: Avid traveler with extensive exploration across **22+ Indian states**; participated in inter-school cultural dance competitions at Methodist High School, Kanpur.

PERSONAL SKILLS & COMPETENCES

- ◇ *Languages:* English, Hindi
- ◇ *General Proficiencies:* Public speaking, teaching, writing, coding, analysis
- ◇ *Hobbies:* Hindi poem reading and writing, Yoga, music, cooking, travel

REFERENCES

- ◇ *Dr. R. Balasubramanian* — Professor (HAG) & Head, Department of Computer Science and Engineering, IIT Roorkee. **Joint Faculty**, Mehta Family School of Data Science and Artificial Intelligence.
Email: bala@cs.iitr.ac.in **Phone:** +91-1332-285852 (O). [*Doctoral Co-Advisor*]
- ◇ *Dr. Sanjeev Kumar* — Professor & Head, Mehta Family School of Data Science and Artificial Intelligence, IIT Roorkee. Professor, Department of Mathematics, IIT Roorkee.
Email: sanjeev.kumar@ma.iitr.ac.in **Phone:** 01332-285824 (O). [*Doctoral Advisor*]

CONTACT

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